Asymptotic growth

Outline

Families of increasing functions of a natural variable

The hierarchy of infinities

Comparison of orders of growth
- Hierarchy of the orders of growth
- Classification of orders of growth
- Properties of logarithmic functions
- Properties of power functions
- Properties of polynomial functions
- Properties of exponential functions

In summary

Introduction (1/2)

- The number of steps executed by an algorithm is a function of a natural number: the **size of its input**
- Two algorithms that solve the same problem are compared by looking at **how fast** these functions grow when the input grows
- Not at their values on a few small inputs

In this chapter we study the functions of a natural variable that appear in the analysis of algorithms, and we compare their **orders of growth** with the asymptotic notations o, $\mathcal{O}$, Θ, Ω and ω.

Introduction (2/2)

The plan of the chapter is the following.

- We introduce the five families of increasing functions of a natural variable used in the course: logarithmic, power, exponential, factorial and self-power functions
- We recall the four limits that put these families in order (the **hierarchy of infinities**)
- We express the comparison of two orders of growth with the asymptotic notation, through the limit criterion; we then derive the hierarchy of the orders of growth, a classification of the orders of growth, and the properties of logarithmic, power, polynomial and exponential functions

Prerequisites

- From this module, the chapter «Asymptotic notation»: the notations o, $\mathcal{O}$, Θ, Ω, ω and the limit criterion
- The functions of a natural variable are the **sequences** of the course *Mathematical Analysis 1* (https://fabiofurini.github.io/mathematical-analysis-1/, Italian version https://fabiofurini.github.io/analisi-matematica-1/)
- Their limits, and the theorems used to compute them, are not explained again: we refer to the part *Limits of sequences* of that course, in particular to the chapters *Computing limits of sequences* (comparison theorem), *Euler's number* and *Hierarchies of infinities and the ratio test*

Families of increasing functions of a natural variable

Introduction: Families of increasing functions

- We define the asymptotically increasing functions of a natural variable
- We introduce the five families used in the course: logarithmic, power, exponential, factorial and self-power functions
- A table and a figure show how differently fast they grow: this is why the order of growth decides which algorithm is faster on large inputs

Functions of a natural variable

- A function of a natural variable is defined on a set of natural numbers of the form

$$\mathbb{N}_{\geq n_0} = \{n_0, n_0 + 1, n_0 + 2, \dots\}$$

- In the analysis of algorithms only its behaviour for **large values of** n matters

- A function is **asymptotically positive** if $f(n) > 0$ for all n from some value onwards

- In the course *Mathematical Analysis 1* such properties are said to hold *eventually*; the following definition describes another property of the same kind

Definition 1: asymptotically increasing function

A function of a natural variable $f : \mathbb{N}_{\geq n_0} \to \mathbb{R}$ *is* ***asymptotically increasing*** *if there exists* $n_1 \in \mathbb{N}_{\geq n_0}$ *such that:*

$$f(n) < f(n+1), \quad \forall n \in \mathbb{N}_{\geq n_1}$$

We now list the principal families
of asymptotically increasing functions
of a natural variable.

The families (1/2)

1. The **logarithmic** functions of a natural variable:

$$f : \mathbb{N}_{\geq 1} \to \mathbb{R}, \quad f : n \mapsto \log_a n \quad \forall a \in \mathbb{R}_{>1}$$

2. The **power** functions of a natural variable:

$$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto n^{\alpha} \quad \forall \alpha \in \mathbb{R}_{>0}$$

3. The **exponential** functions of a natural variable:

$$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto b^n \quad \forall b \in \mathbb{R}_{>1}$$

The families (2/2)

4. The **factorial** function of a natural variable:

$$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto n!$$

5. The **self-power** function of a natural variable:

$$f : \mathbb{N} \to \mathbb{R}, \quad f : n \mapsto n^n$$

where, by convention, we set $0^0 = 1$.

Properties of the five families

- All these functions are asymptotically positive, and all of them diverge to $+\infty$ as $n \to +\infty$
- The factorial and the self-power functions are not increasing on their whole domain, since

$$0! = 1! = 1 \qquad \text{and} \qquad 0^0 = 1^1 = 1,$$

  but they are strictly increasing from $n = 1$ onwards: this is why we speak of *asymptotically* increasing functions
- In these slides $\ln n = \log_e n$ denotes the natural logarithm, where e is Euler's number

How different are these functions?
We look at their values
for larger and larger values of n.

Example (1/4)

- We compute the values of some increasing functions of a natural variable:

$$\ln n, \qquad n^2, \qquad 2^n, \qquad n!, \qquad n^n$$

- We increase n by one order of magnitude at a time, from 1 to 100 million
- The values of $\ln n$ and the leading factors of 2^n and $n!$ are rounded to one decimal digit
- The values of n^2 and n^n are exact

Example (2/4)

n	$\ln n$	n^2	2^n
$1 = 10^0$	0	1	2
$10 = 10^1$	≈ 2.3	10^2	$\approx 1.0 \cdot 10^3$
$100 = 10^2$	≈ 4.6	10^4	$\approx 1.3 \cdot 10^{30}$
$1{,}000 = 10^3$	≈ 6.9	10^6	$\approx 1.1 \cdot 10^{301}$
$10{,}000 = 10^4$	≈ 9.2	10^8	$\approx 2.0 \cdot 10^{3{,}010}$
$100{,}000 = 10^5$	≈ 11.5	10^{10}	$\approx 1.0 \cdot 10^{30{,}103}$
10^6	≈ 13.8	10^{12}	$\approx 9.9 \cdot 10^{301{,}029}$
10^7	≈ 16.1	10^{14}	$\approx 9.0 \cdot 10^{3{,}010{,}299}$
10^8	≈ 18.4	10^{16}	$\approx 3.7 \cdot 10^{30{,}102{,}999}$

Example (3/4)

n	$n!$	n^n
$1 = 10^0$	1	1
$10 = 10^1$	$\approx 3.6 \cdot 10^6$	10^{10}
$100 = 10^2$	$\approx 9.3 \cdot 10^{157}$	10^{200}
$1{,}000 = 10^3$	$\approx 4.0 \cdot 10^{2{,}567}$	$10^{3{,}000}$
$10{,}000 = 10^4$	$\approx 2.8 \cdot 10^{35{,}659}$	$10^{40{,}000}$
$100{,}000 = 10^5$	$\approx 2.8 \cdot 10^{456{,}573}$	$10^{500{,}000}$
10^6	$\approx 8.3 \cdot 10^{5{,}565{,}708}$	$10^{6{,}000{,}000}$
10^7	$\approx 1.2 \cdot 10^{65{,}657{,}059}$	$10^{70{,}000{,}000}$
10^8	$\approx 1.6 \cdot 10^{756{,}570{,}556}$	$10^{800{,}000{,}000}$

Example (4/4)

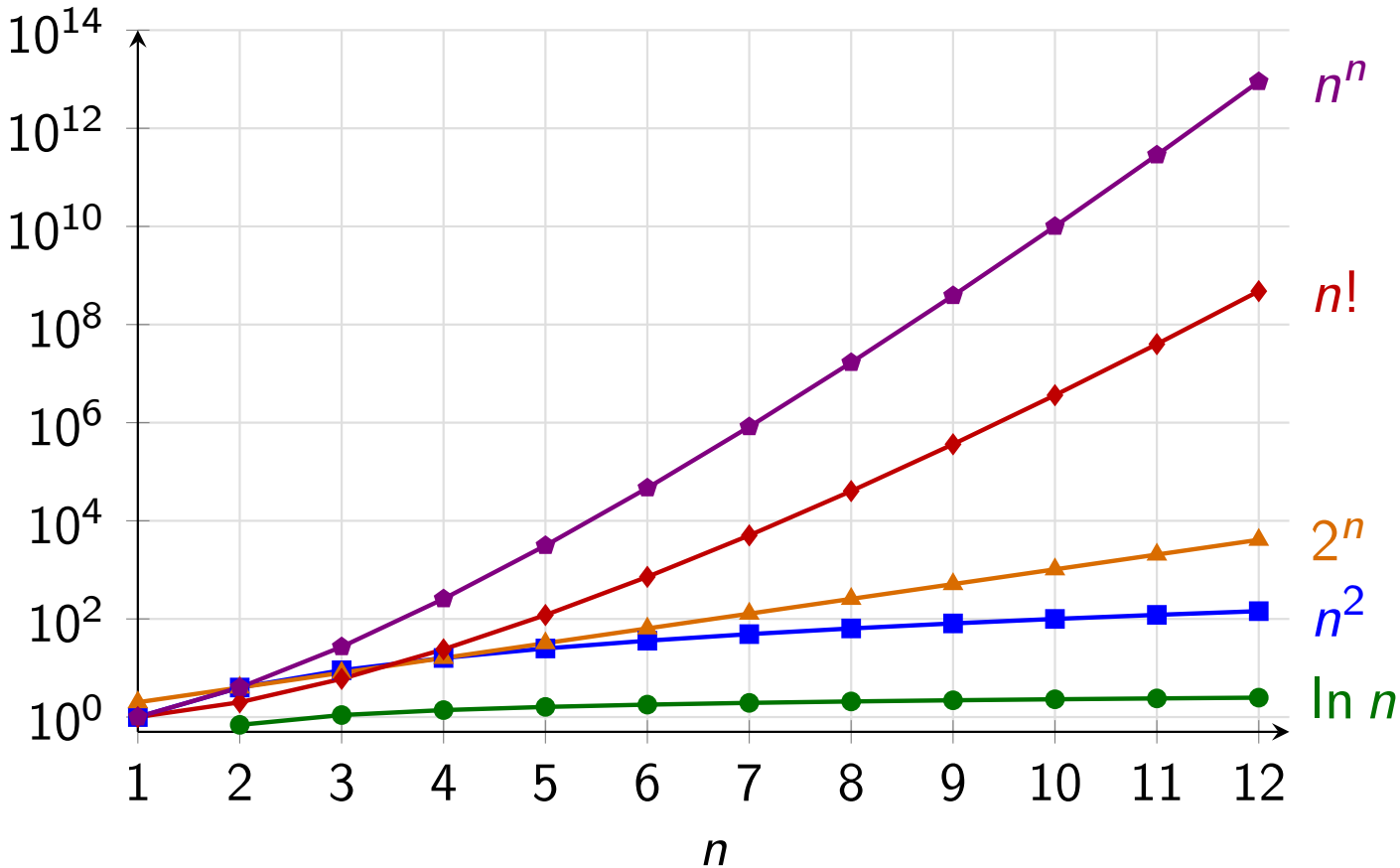
- To appreciate these numbers: the number of atoms in the observable universe is estimated at about 10^{82}
- A personal computer of 2024 executes about $10 \cdot 10^9 = 10^{10}$ operations per second (10 GFLOPS)
- An algorithm that executes 2^n operations on an input with $n = 100$ needs

$$\frac{2^{100}}{10^{10}} \text{ seconds} \approx 1.3 \cdot 10^{20} \text{ seconds} \approx 4.0 \cdot 10^{12} \text{ years}$$

- An algorithm that executes n^2 operations needs $10^4/10^{10} = 10^{-6}$ seconds

The first values

- The figure shows the first values of the same functions; the vertical axis is in **logarithmic scale**: each horizontal line of the grid is a multiplication by 100



Small values of n

- The figure shows $\ln n$, n^2, 2^n, $n!$ and n^n for $n \in \{1, 2, \ldots, 12\}$; the points are joined by segments only for readability
- For small values of n the order of the functions can be **different from the final one**:

$$2^3 = 8 < 9 = 3^2, \qquad \text{and} \qquad 2^n > n! \quad \text{for } n \in \{1, 2, 3\}$$

- This is why we compare the functions only for large values of n

Summary

- A function of a natural variable is **asymptotically increasing** if $f(n) < f(n+1)$ from some n_1 onwards
- The five families: **logarithmic** $\log_a n$, **power** n^α, **exponential** b^n, **factorial** $n!$ and **self-power** n^n, with $a, b > 1$ and $\alpha > 0$
- All of them diverge to $+\infty$, but at very different speeds: with $n = 100$, n^2 operations take 10^{-6} seconds, 2^n operations about $4.0 \cdot 10^{12}$ years

The hierarchy of infinities

Introduction: The hierarchy of infinities

- The five families all diverge to $+\infty$: to decide which one diverges faster, we compute the **limit of the ratio** of two of them
- These limits are the *hierarchy of infinities* for sequences of the course *Mathematical Analysis 1* (chapter *Hierarchies of infinities and the ratio test*), where they are proved
- Here we recall them, with the idea of their proofs, and we extend the first one to the powers of the logarithms

Polylogarithmic functions

- Given $a \in \mathbb{R}_{>1}$ and $\beta \in \mathbb{R}_{>0}$, the asymptotically increasing function of a natural variable

$$f : \mathbb{N}_{\geq 1} \to \mathbb{R}, \quad f : n \mapsto \log_a^\beta n = \left(\log_a n\right)^\beta$$

  is called a **polylogarithmic** function of a natural variable

- For $\beta = 1$ it is a logarithmic function

Proposition 1: logarithms and powers

For every $\alpha \in \mathbb{R}_{>0}$, $\beta \in \mathbb{R}_{>0}$ *and* $a \in \mathbb{R}_{>1}$, *we have:*

$$\lim_{n \to +\infty} \frac{\log_a^\beta n}{n^\alpha} = 0^+$$

- Idea of the proof: the case $\beta = 1$ comes from the course *Mathematical Analysis 1*, and the general case is reduced to it

Proof (1/2)

- For $\beta = 1$ this is the hierarchy of infinities (part I) of the course *Mathematical Analysis 1*:

$$\lim_{n \to +\infty} \frac{\log_a n}{n^\alpha} = 0^+, \quad \forall \alpha \in \mathbb{R}_{>0}, \ \forall a \in \mathbb{R}_{>1}$$

- It is proved with the inequality

$$\log_a x < x \log_a 2, \qquad \forall x \in \mathbb{R}_{>0}$$

(a consequence of Bernoulli's inequality), applied to $x = n^{\alpha/2}$, and with the comparison theorem

Proof (2/2)

- For every $\beta \in \mathbb{R}_{>0}$, since $\frac{\alpha}{\beta} \in \mathbb{R}_{>0}$, we then have:

$$\lim_{n \to +\infty} \frac{\log_a^\beta n}{n^\alpha} = \lim_{n \to +\infty} \left(\frac{\log_a n}{\underbrace{n^{\alpha/\beta}}_{\to 0^+}} \right)^\beta = 0^+$$

□

Proposition 2: powers and exponentials

For every $\alpha \in \mathbb{R}_{>0}$ *and* $b \in \mathbb{R}_{>1}$, *we have:*

$$\lim_{n \to +\infty} \frac{n^\alpha}{b^n} = 0^+$$

- This is the hierarchy of infinities (part II)

Powers and exponentials: idea of the proof

- It follows from the Proposition «logarithms and powers» by replacing n with 2^n, which gives

$$\frac{n}{(2^\mu)^n} \to 0^+, \qquad \forall \mu \in \mathbb{R}_{>0}$$

- Hence $\frac{n}{b^n} \to 0^+$ for every $b \in \mathbb{R}_{>1}$ (choose μ such that $2^\mu = b$)

- The general case follows from the identity

$$\frac{n^\alpha}{b^n} = \left(\frac{n}{\left(b^{1/\alpha}\right)^n}\right)^\alpha, \qquad b^{1/\alpha} \in \mathbb{R}_{>1}$$

Proposition 3: exponentials and factorial

For every $b \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{b^n}{n!} = 0^+$$

- This is the hierarchy of infinities (part III). It follows from the **ratio test**, since

$$\frac{b^{n+1}}{(n+1)!} \frac{n!}{b^n} = \frac{b}{n+1} \to 0^+ < 1$$

Proposition 4: factorial and self-power

$$\lim_{n \to +\infty} \frac{n!}{n^n} = 0^+$$

- This is the hierarchy of infinities (part IV). It follows from the **ratio test**, since

$$\frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \to \frac{1}{e} < 1$$

The order of the five families

Polylogarithmic functions (with base greater than one)
diverge more slowly than
power functions (with positive exponent),
which diverge more slowly than
exponential functions (with base greater than one),
which diverge more slowly than
the **factorial** function,
which diverges more slowly than
the **self-power** function.

Summary

With $a, b \in \mathbb{R}_{>1}$ and $\alpha, \beta \in \mathbb{R}_{>0}$:

$$\lim_{n \to +\infty} \frac{\log_a^\beta n}{n^\alpha} = 0^+, \qquad \lim_{n \to +\infty} \frac{n^\alpha}{b^n} = 0^+,$$

$$\lim_{n \to +\infty} \frac{b^n}{n!} = 0^+, \qquad \lim_{n \to +\infty} \frac{n!}{n^n} = 0^+$$

- The first two come from the hierarchy of infinities of *Mathematical Analysis 1* (Bernoulli's inequality and the comparison theorem)
- The last two follow from the **ratio test**
- Each family diverges more slowly than the next one

Comparison of orders of growth

Introduction: Comparison of orders of growth

- We express the comparison of two orders of growth with the asymptotic notation, through the **limit criterion**
- We translate the hierarchy of infinities into the **hierarchy of the orders of growth**, and we classify orders of growth
- We prove the properties of logarithmic, power, polynomial and exponential functions that are used in the analysis of algorithms

Order of growth

- The **order of growth** of an asymptotically increasing function of a natural variable describes how rapidly its values increase compared to the values of another asymptotically increasing function of a natural variable

- We compare orders of growth with the asymptotic notations o, $\mathcal{O}$, Θ, Ω and ω of the chapter «Asymptotic notation»

- For the functions of this chapter the ratio $\frac{f(n)}{g(n)}$ always has a limit, and the notations that hold are decided by the **limit criterion** proved there, which we recall

Proposition 5: limit criterion

Let f and g be two asymptotically positive functions of a natural variable such that the limit $L = \lim_{n\to+\infty} \frac{f(n)}{g(n)}$ exists ($L = 0^+$, $L \in \mathbb{R}_{>0}$ or $L = +\infty$). Then:

$$
\begin{aligned}
f(n) = o\big(g(n)\big) &\iff L = 0^+, \\
f(n) = \mathcal{O}\big(g(n)\big) &\iff L = 0^+ \text{ or } L \in \mathbb{R}_{>0}, \\
f(n) = \Theta\big(g(n)\big) &\iff L \in \mathbb{R}_{>0}, \\
f(n) = \Omega\big(g(n)\big) &\iff L \in \mathbb{R}_{>0} \text{ or } L = +\infty, \\
f(n) = \omega\big(g(n)\big) &\iff L = +\infty.
\end{aligned}
$$

How we use the limit criterion

- The proof is in the chapter «Asymptotic notation» (Proposition «limit criterion»), where the five notations are defined through positive constants and a threshold
- In this chapter we **compute the limit of the ratio** and apply the limit criterion

Reading the notations (1/2)

Given two asymptotically increasing functions of a natural variable f and g:

- if $f(n) = o\big(g(n)\big)$, f has a strictly **smaller order of growth** than g
  $\rightsquigarrow$ the function f **grows slower than** the function g;
- if $f(n) = \mathcal{O}\big(g(n)\big)$, f has a **smaller or equal order of growth** compared to g
  $\rightsquigarrow$ the function f **grows at most as fast as** the function g;
- if $f(n) = \Theta\big(g(n)\big)$, f has the **same order of growth** as g
  $\rightsquigarrow$ the function f **grows as fast as** the function g;

Reading the notations (2/2)

- if $f(n) = \Omega\big(g(n)\big)$, f has a **greater or equal order of growth** compared to g
  $\rightsquigarrow$ the function f **grows at least as fast as** the function g;

- if $f(n) = \omega\big(g(n)\big)$, f has a strictly **greater order of growth** than g
  $\rightsquigarrow$ the function f **grows faster than** the function g.

Properties of the notations (1/2)

- The notations do not depend on the values of f and g on **finitely many** natural numbers
- Changing them changes neither the limit of the ratio nor, with a larger threshold, the inequalities of the definitions

Properties of the notations (2/2)

- By the relational properties of the chapter «Asymptotic notation» (symmetry and transpose symmetry), exchanging the two functions exchanges o with ω and $\mathcal{O}$ with Ω, while Θ does not change:

$$f(n) = o\big(g(n)\big) \iff g(n) = \omega\big(f(n)\big)$$
$$f(n) = \Theta\big(g(n)\big) \iff g(n) = \Theta\big(f(n)\big)$$

- All five notations are also **transitive**: for example, $f(n) = o\big(g(n)\big)$ and $g(n) = o\big(h(n)\big)$ give $f(n) = o\big(h(n)\big)$

We now compare three pairs of functions
by computing the limit of their ratio.

Example (1/4)

- We have $\ln n = o\left(\sqrt{n}\right)$, since by the Proposition «logarithms and powers» (with $a = e$, $\beta = 1$, $\alpha = \frac{1}{2}$):

$$\lim_{n \to +\infty} \frac{\ln n}{\sqrt{n}} = 0^+$$

- We have $2n^2 + 3n = \Theta(n^2)$, since:

$$\lim_{n \to +\infty} \frac{2n^2 + 3n}{n^2} = \lim_{n \to +\infty} \left(2 + \underbrace{\frac{3}{n}}_{\to 0}\right) = 2 \in \mathbb{R}_{>0}$$

Example (2/4)

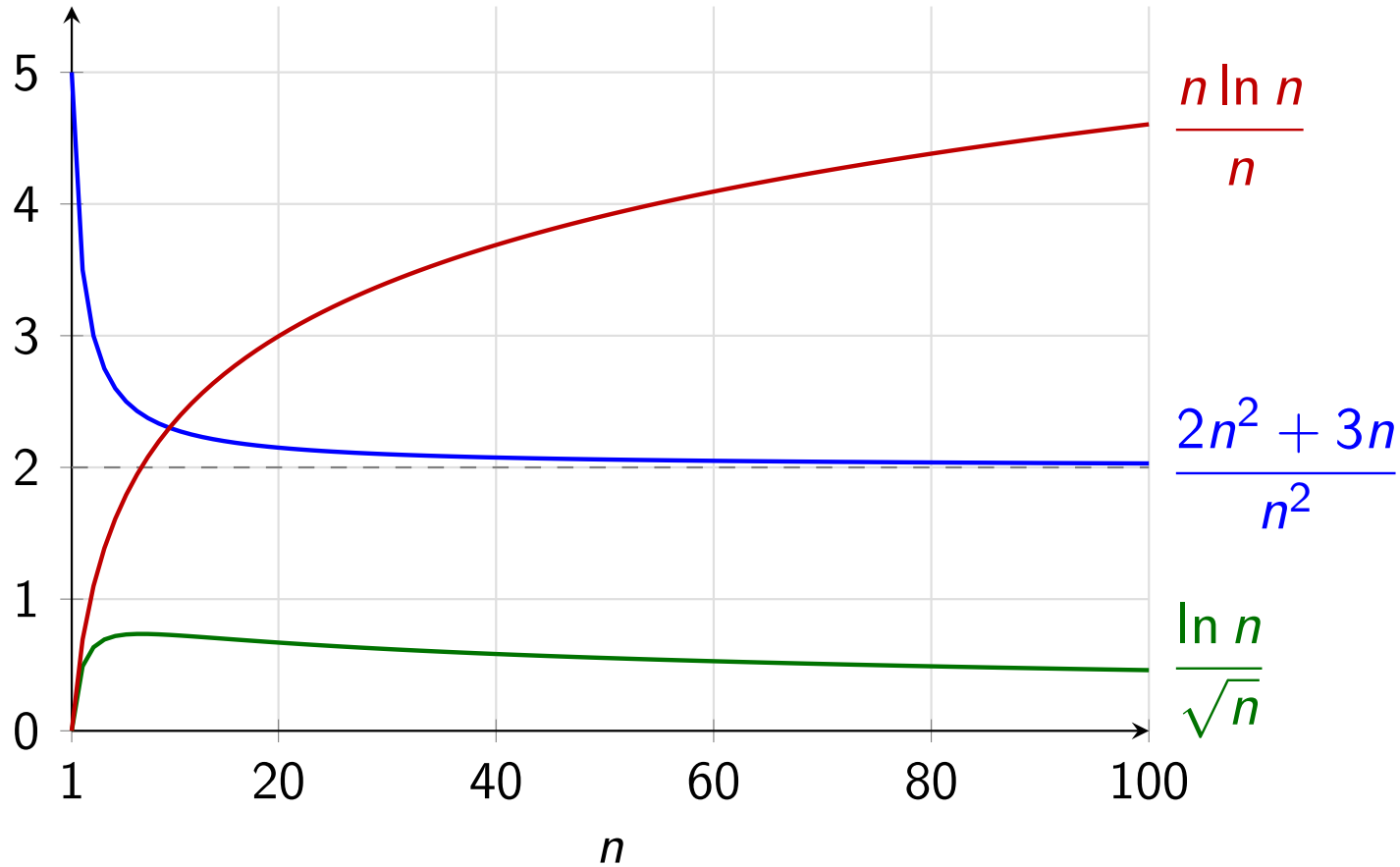
- We have $n \ln n = \omega(n)$, since:

$$\lim_{n \to +\infty} \frac{n \ln n}{n} = \lim_{n \to +\infty} \ln n = +\infty$$

- The three ratios are drawn in the next figure

- The first ratio is **not monotone**: it increases up to its largest value, about 0.74, reached for $n = 7$, and only then it decreases towards 0

Example (3/4)


5
4
3
2
1
0
1
20
40
60
80
100
n
$\frac{n \ln n}{n}$
$\frac{2n^2 + 3n}{n^2}$
$\frac{\ln n}{\sqrt{n}}$

Example (4/4)

- The figure shows the three ratios for $n \in \{1, 2, \ldots, 100\}$ (the points are joined by segments)
- The first one tends to 0: $\ln n = o(\sqrt{n})$
- The second one tends to 2 (dashed line): $2n^2 + 3n = \Theta(n^2)$
- The third one diverges to $+\infty$: $n \ln n = \omega(n)$

When the limit does not exist

- The limit criterion requires the limit of the ratio to **exist**
- For example, for $f(n) = n$ if n is even and $f(n) = 2n$ if n is odd, the ratio $\frac{f(n)}{n}$ takes alternately the values 1 and 2 and has no limit
- Nevertheless $f(n) = \Theta(n)$, by the definition of the chapter «Asymptotic notation» with $c_1 = 1$, $c_2 = 2$ and $n_1 = 0$, since

$$0 \leq n \leq f(n) \leq 2n, \qquad \forall n \in \mathbb{N}$$

- All the comparisons of this chapter are made between functions whose ratio has a limit

Summary

- The **limit criterion**: if $\frac{f(n)}{g(n)}$ tends to 0^+, to a positive real number or to $+\infty$, then $f(n) = o\big(g(n)\big)$, $f(n) = \Theta\big(g(n)\big)$ or $f(n) = \omega\big(g(n)\big)$
- o, $\mathcal{O}$, Θ, Ω, ω read as: grows slower, at most as fast, as fast, at least as fast, faster
- The notations ignore finitely many values, are transitive, and exchanging the functions exchanges o with ω and $\mathcal{O}$ with Ω
- If the limit does not exist, the criterion says nothing: we go back to the definitions

Hierarchy of the orders of growth

- By the limit criterion, the limits of the hierarchy of infinities translate directly into relations o between the families
- We obtain four observations, one for each pair of consecutive families, and a few examples

- Any polylogarithmic function of a natural variable with a base strictly greater than one is o of any power function of a natural variable with a positive exponent

Observation 1: polylogarithms and powers

For every $\alpha \in \mathbb{R}_{>0}$, $\beta \in \mathbb{R}_{>0}$ *and* $a \in \mathbb{R}_{>1}$, *we have* $\log_a^\beta n = o(n^\alpha)$.

Proof

- By the Proposition «logarithms and powers», for every $\alpha \in \mathbb{R}_{>0}$, $\beta \in \mathbb{R}_{>0}$ and $a \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{\log_a^\beta n}{n^\alpha} = 0^+$$

- The claim follows from the limit criterion

□

- Any power function of a natural variable with a positive exponent is o of any exponential function of a natural variable with a base strictly greater than one

Observation 2: powers and exponentials

For every $\alpha \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>1}$, we have $n^\alpha = o(b^n)$.

Proof

- By the Proposition «powers and exponentials», for every $\alpha \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{n^\alpha}{b^n} = 0^+$$

- The claim follows from the limit criterion

□

With these two observations
we can compare sums of many terms:
each term is divided by the same power.

Example (1/3)

- We have $\ln^6 n + 2 + 2^{1/n} + 3^{-n} + 9\,n^2 = o(n^3)$, since:

$$\lim_{n \to +\infty} \frac{\ln^6 n + 2 + 2^{1/n} + 3^{-n} + 9\,n^2}{n^3}$$

$$= \lim_{n \to +\infty} \left(\underbrace{\frac{\ln^6 n}{n^3}}_{\to 0^+} + \underbrace{\frac{2}{n^3}}_{\to 0^+} + \underbrace{\frac{2^{1/n}}{n^3}}_{\to 0^+} + \underbrace{\frac{3^{-n}}{n^3}}_{\to 0^+} + \underbrace{\frac{9}{n}}_{\to 0^+} \right) = 0^+$$

Example (2/3)

- We have $3n^2 + \sqrt{n} + \ln n + 5 + 4^{-n} + 10\, n^{1/3} = o(n^3)$, since:

$$\lim_{n \to +\infty} \frac{3n^2 + \sqrt{n} + \ln n + 5 + 4^{-n} + 10\, n^{1/3}}{n^3}$$

$$= \lim_{n \to +\infty} \left(\underbrace{\frac{3}{n}}_{\to 0^+} + \underbrace{\frac{1}{n^{5/2}}}_{\to 0^+} + \underbrace{\frac{\ln n}{n^3}}_{\to 0^+} + \underbrace{\frac{5}{n^3}}_{\to 0^+} + \underbrace{\frac{4^{-n}}{n^3}}_{\to 0^+} + \underbrace{\frac{10}{n^{8/3}}}_{\to 0^+} \right) = 0^+$$

Example (3/3)

- We have $n^{1/4} + \ln^2 n + \frac{7}{\sqrt{n}} + e^{-n} + 20\, n^{1/2} = o(n^2)$, since:

$$\lim_{n \to +\infty} \frac{n^{1/4} + \ln^2 n + \frac{7}{\sqrt{n}} + e^{-n} + 20\, n^{1/2}}{n^2}$$

$$= \lim_{n \to +\infty} \left(\underbrace{\frac{1}{n^{7/4}}}_{\to 0^+} + \underbrace{\frac{\ln^2 n}{n^2}}_{\to 0^+} + \underbrace{\frac{7}{n^{5/2}}}_{\to 0^+} + \underbrace{\frac{e^{-n}}{n^2}}_{\to 0^+} + \underbrace{\frac{20}{n^{3/2}}}_{\to 0^+} \right) = 0^+$$

- In each case the terms with a logarithm go to 0^+ by the Proposition «logarithms and powers», the other terms by elementary limits, and the claim follows from the limit criterion

- Any exponential function of a natural variable with a base strictly greater than one is o of the factorial function of a natural variable

Observation 3: exponentials and factorial

For every $b \in \mathbb{R}_{>1}$, we have $b^n = o(n!)$.

Proof

- By the Proposition «exponentials and factorial», for every $b \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{b^n}{n!} = 0^+$$

- The claim follows from the limit criterion

□

- The factorial function of a natural variable is o of the self-power function of a natural variable

Observation 4: factorial and self-power

We have $n! = o(n^n)$.

Proof

- By the Proposition «factorial and self-power», we have:

$$\lim_{n \to +\infty} \frac{n!}{n^n} = 0^+$$

- The claim follows from the limit criterion

□

Example (1/2)

- We have $n! + 2^n + \sqrt{n} + \ln^3 n + n^{n-1} + \frac{1}{n} = o(n^n)$, since:

$$\lim_{n \to +\infty} \frac{n! + 2^n + \sqrt{n} + \ln^3 n + n^{n-1} + \frac{1}{n}}{n^n}$$

$$= \lim_{n \to +\infty} \left(\underbrace{\frac{n!}{n^n}}_{\to 0^+} + \underbrace{\frac{2^n}{n^n}}_{\to 0^+} + \underbrace{\frac{\sqrt{n}}{n^n}}_{\to 0^+} + \underbrace{\frac{\ln^3 n}{n^n}}_{\to 0^+} + \underbrace{\frac{1}{n}}_{\to 0^+} + \underbrace{\frac{1}{n^{n+1}}}_{\to 0^+} \right) = 0^+$$

Example (2/2)

- For example, the second term

$$\frac{2^n}{n^n} = \frac{2^n}{n!}\frac{n!}{n^n}$$

is the product of two functions that tend to 0^+, by the Propositions «exponentials and factorial» and «factorial and self-power»

Hierarchy of the orders of growth

- For every $a \in \mathbb{R}_{>1}$, $\beta \in \mathbb{R}_{>0}$, $\alpha \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>1}$, we have

$$\log_a^\beta n = o(n^\alpha), \quad n^\alpha = o(b^n), \quad b^n = o(n!), \quad n! = o(n^n)$$

- By the transitivity of o (chapter «Asymptotic notation»), each function of the chain is o of all the functions that follow it, for example

$$\log_a^\beta n = o(n!)$$

Summary

- Each limit of the hierarchy of infinities, with the limit criterion, gives a relation o:

$$\log_a^\beta n = o(n^\alpha), \quad n^\alpha = o(b^n), \quad b^n = o(n!), \quad n! = o(n^n)$$

- By transitivity, each function of the chain is o of all the following ones

- To compare a sum with a function g, we divide **each term** by $g(n)$ and check that every ratio tends to 0^+

Classification of orders of growth

The families of the hierarchy are used as **yardsticks**
to classify the order of growth
of any other function.

Definition 2: bounded orders of growth

An asymptotically increasing function of a natural variable f has:

- *a **polylogarithmically bounded order of growth**, if there exists $\beta \in \mathbb{R}_{>0}$ such that $f(n) = \mathcal{O}(\ln^{\beta} n)$; and a **logarithmically bounded order of growth**, if $f(n) = \mathcal{O}(\ln n)$;*
- *a **polynomially bounded order of growth**, if there exists $\alpha \in \mathbb{R}_{>0}$ such that $f(n) = \mathcal{O}(n^{\alpha})$;*
- *an **exponentially bounded order of growth**, if there exists $b \in \mathbb{R}_{>1}$ such that $f(n) = \mathcal{O}(b^{n})$;*
- *a **factorially bounded order of growth**, if $f(n) = \mathcal{O}(n!)$.*

Definition 3: linear, sub-linear and super-linear order of growth

An asymptotically increasing function of a natural variable f has:

- *a* ***sub-linear order of growth***, *if* $f(n) = o(n)$;
- *a* ***linear order of growth***, *if* $f(n) = \Theta(n)$;
- *a* ***super-linear order of growth***, *if* $f(n) = \omega(n)$.

The classes are nested

- By the hierarchy of the orders of growth, a function with a polylogarithmically bounded order of growth has also a polynomially bounded order of growth
- which in turn is also exponentially bounded and factorially bounded
- For example, $f(n) = \mathcal{O}(\ln^{\beta} n)$ and $\ln^{\beta} n = o(n)$ give $f(n) = \mathcal{O}(n)$, since o implies $\mathcal{O}$ and $\mathcal{O}$ is transitive (chapter «Asymptotic notation»)

We now classify four functions:
$\sqrt{n}$, n, $n \ln n$ and n^2.

Example (1/4)

- $\sqrt{n} = o(n)$, since

$$\frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} \to 0^+$$

  a **sub-linear** order of growth

- It has a polynomially bounded order of growth, since $\sqrt{n} = \mathcal{O}(n^{1/2})$

Example (2/4)

- But $\sqrt{n}$ has **not** a polylogarithmically bounded order of growth:
- since $\frac{\ln^\beta n}{\sqrt{n}} \to 0^+$ (Proposition «logarithms and powers»), we have

$$\frac{\sqrt{n}}{\ln^\beta n} \to +\infty$$

- that is, by the limit criterion,

$$\sqrt{n} = \omega(\ln^\beta n) \qquad \text{and} \qquad \sqrt{n} \neq \mathcal{O}(\ln^\beta n), \qquad \forall \beta \in \mathbb{R}_{>0}$$

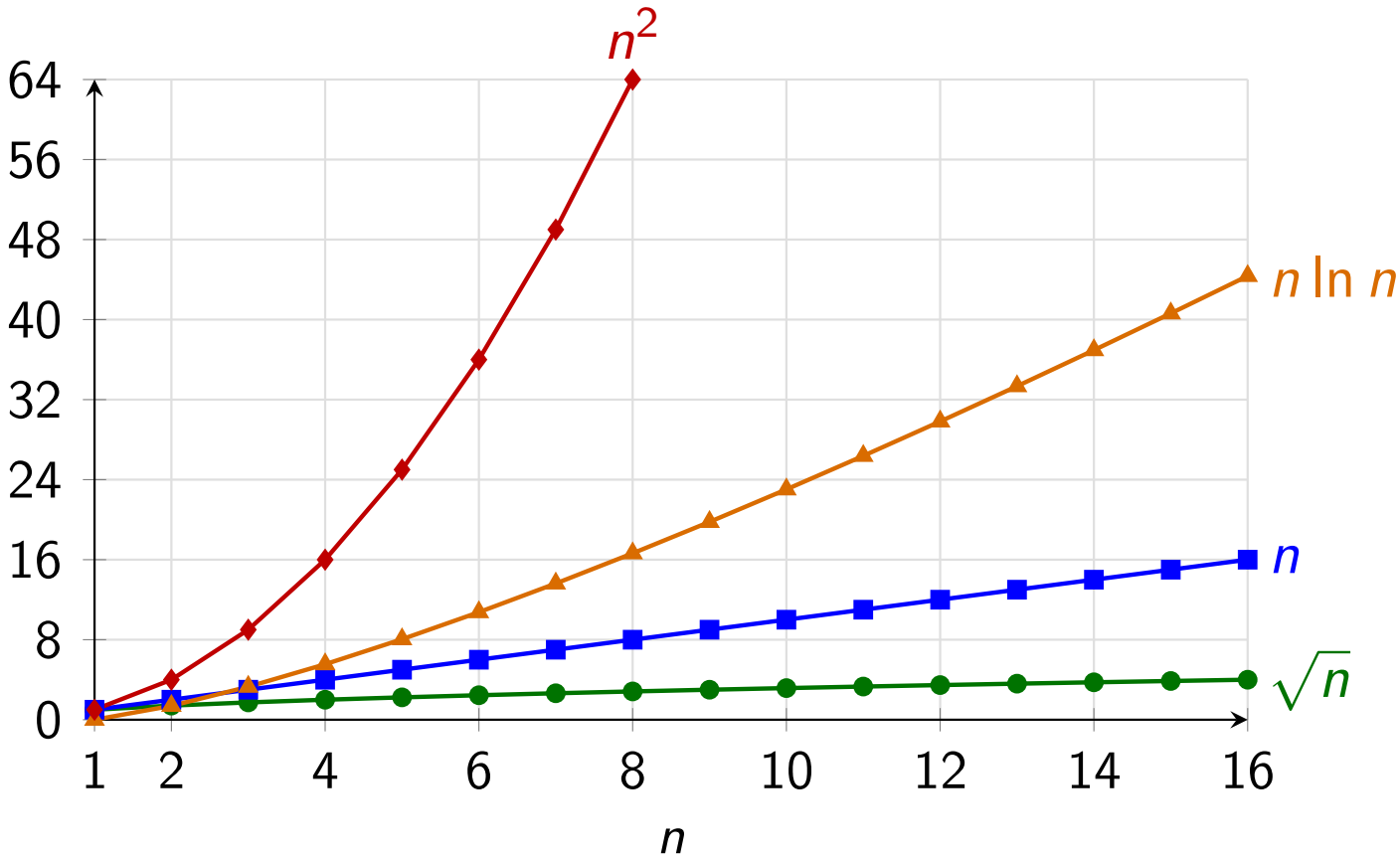
Example (3/4)

- $n = \Theta(n)$: a **linear** order of growth
- $n \ln n = \omega(n)$, since $\frac{n \ln n}{n} = \ln n \to +\infty$: a **super-linear** order of growth
- It has a polynomially bounded order of growth, since

$$\frac{n \ln n}{n^2} = \frac{\ln n}{n} \to 0^+,$$

  so that $n \ln n = o(n^2)$ and $n \ln n = \mathcal{O}(n^2)$
- $n^2 = \omega(n)$, since $\frac{n^2}{n} = n \to +\infty$: a **super-linear** order of growth; it has a polynomially bounded order of growth, since $n^2 = \mathcal{O}(n^2)$

Example (4/4)



A sub-linear function ($\sqrt{n}$), the linear function n and two super-linear functions ($n \ln n$ and n^2) for $n \in \{1, 2, \ldots, 16\}$ (n^2 only up to $n = 8$, where it reaches 64).

Summary

- An order of growth is **polylogarithmically**, **polynomially**, **exponentially** or **factorially bounded** if $f(n) = \mathcal{O}(\ln^\beta n)$, $\mathcal{O}(n^\alpha)$, $\mathcal{O}(b^n)$ or $\mathcal{O}(n!)$; each class is contained in the following ones
- It is **sub-linear**, **linear** or **super-linear** if $f(n) = o(n)$, $\Theta(n)$ or $\omega(n)$
- $\sqrt{n}$ is sub-linear and polynomially bounded, but not polylogarithmically bounded; $n \ln n$ and n^2 are super-linear and polynomially bounded

Properties of logarithmic functions

- We now prove the properties of logarithmic, power, polynomial and exponential functions that are used in the analysis of algorithms
- In all the proofs we compute the **limit of the ratio** of the two functions and we apply the limit criterion

- Any two logarithmic functions of a natural variable with bases strictly greater than one are Θ of each other; multiplying the argument by a positive constant does not change the order of growth either

Observation 5: change of base

For every $a \in \mathbb{R}_{>1}$, $c \in \mathbb{R}_{>1}$ *and* $\lambda \in \mathbb{R}_{>0}$, *we have* $\log_a(\lambda\, n) = \Theta(\log_c n)$.

Proof

- For every $a, c \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, we use

$$\log_a(\lambda\, n) = \log_a \lambda + \log_a n \qquad \text{and} \qquad \log_a n = \frac{\log_c n}{\log_c a}$$

- We have:

$$\lim_{n\to+\infty} \frac{\log_a(\lambda\, n)}{\log_c n} = \lim_{n\to+\infty} \left(\underbrace{\frac{\log_a \lambda}{\log_c n}}_{\to 0} + \frac{\frac{\log_c n}{\log_c a}}{\log_c n} \right) = \frac{1}{\log_c a} \in \mathbb{R}_{>0}$$



The base of a logarithm

- For this reason, when only the order of growth matters, the **base of a logarithm is often omitted**
- For example, $\log_2 n = \Theta(\ln n)$, and

$$\lim_{n \to +\infty} \frac{\log_2 n}{\ln n} = \frac{1}{\ln 2} \approx 1.44$$

- Adding a positive constant to the argument of a logarithmic function of a natural variable does not change its order of growth

Observation 6: shift of the argument

For every $a \in \mathbb{R}_{>1}$ *and* $\lambda \in \mathbb{R}_{>0}$, *we have* $\log_a(n + \lambda) = \Theta(\log_a n)$.

Proof

For every $a \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{\log_a(n+\lambda)}{\log_a n} = \lim_{n \to +\infty} \frac{\log_a\left(n\left(1+\frac{\lambda}{n}\right)\right)}{\log_a n}$$

$$= \lim_{n \to +\infty} \frac{\log_a n + \overbrace{\log_a\left(1+\frac{\lambda}{n}\right)}^{\to 0}}{\log_a n}$$

$$= 1$$

□

- The **quasi-linear** function of a natural variable $n \mapsto n \log_a n$ is o of any power function of a natural variable with exponent strictly greater than one

Observation 7: quasi-linear and powers

For every $\alpha \in \mathbb{R}_{>1}$ and $a \in \mathbb{R}_{>1}$, we have $n \log_a n = o(n^\alpha)$.

Proof

- For every $\alpha \in \mathbb{R}_{>1}$ and $a \in \mathbb{R}_{>1}$, we have $\alpha - 1 \in \mathbb{R}_{>0}$
- By the Proposition «logarithms and powers»:

$$\lim_{n \to +\infty} \frac{n \log_a n}{n^\alpha} = \lim_{n \to +\infty} \frac{\log_a n}{n^{\alpha - 1}} = 0^+$$

□

- The quasi-linear function of a natural variable $n \mapsto n \log_a n$ is ω of the linear function of a natural variable: it has a super-linear order of growth

Observation 8: quasi-linear and linear

For every $a \in \mathbb{R}_{>1}$, we have $n \log_a n = \omega(n)$.

Proof

For every $a \in \mathbb{R}_{>1}$, we have:

$$\lim_{n \to +\infty} \frac{n \log_a n}{n} = \lim_{n \to +\infty} \log_a n = +\infty$$

□

Summary

- The **base** of a logarithm and a **constant factor** in its argument do not change the order of growth: $\log_a(\lambda\, n) = \Theta(\log_c n)$
- A **constant added** to the argument does not change it either: $\log_a(n + \lambda) = \Theta(\log_a n)$
- The quasi-linear function $n \log_a n$ lies strictly between n and every power with exponent greater than one:

$$n \log_a n = \omega(n), \qquad n \log_a n = o(n^\alpha) \quad \forall \alpha \in \mathbb{R}_{>1}$$

Properties of power functions

- We see which changes of the argument leave the order of growth of a power function unchanged
- We compare the power functions with the linear function

- Adding a positive constant to the argument of a power function of a natural variable does not change its order of growth

Observation 9: shift of the argument

For every $\alpha \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{>0}$, we have $(n+\lambda)^\alpha = \Theta(n^\alpha)$.

Proof

For every $\alpha \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{(n+\lambda)^\alpha}{n^\alpha} = \lim_{n \to +\infty} \left(1 + \underbrace{\frac{\lambda}{n}}_{\to 0}\right)^\alpha = 1$$

□

- Multiplying the argument of a power function of a natural variable by a positive constant does not change its order of growth

Observation 10: scaling of the argument

For every $\alpha \in \mathbb{R}_{>0}$ *and* $\lambda \in \mathbb{R}_{>0}$, *we have* $(\lambda\, n)^{\alpha} = \Theta(n^{\alpha})$.

Proof

For every $\alpha \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{(\lambda\, n)^\alpha}{n^\alpha} = \lim_{n \to +\infty} \frac{\lambda^\alpha\, n^\alpha}{n^\alpha} = \lambda^\alpha \in \mathbb{R}_{>0}$$

□

- The order of growth of a power function of a natural variable, compared to the linear function of a natural variable, depends on whether the exponent is greater or smaller than one

Observation 11: powers and the linear function

For every $\alpha \in \mathbb{R}_{>0}$, we have:

- $n^\alpha = \omega(n)$ *(super-linear order of growth), if* $\alpha > 1$;
- $n^\alpha = o(n)$ *(sub-linear order of growth), if* $\alpha < 1$.

Proof

For every $\alpha \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{n^\alpha}{n} = \lim_{n \to +\infty} n^{\alpha - 1} = \begin{cases} +\infty & \text{if} \quad \alpha > 1 \\ 0^+ & \text{if} \quad \alpha < 1 \end{cases}$$

□

- Any polynomial function of a natural variable of degree k with a positive leading coefficient is $\Theta(n^k)$: it has the same order of growth as its **leading term**

Observation 12: polynomials

For every $k \in \mathbb{N}_{>1}$ *and every* $a_0, a_1, \ldots, a_k \in \mathbb{R}$ *with* $a_k \in \mathbb{R}_{>0}$, *we have* $\sum_{i=0}^{k} a_i \, n^i = \Theta(n^k)$.

Proof

We have:

$$\lim_{n\to+\infty} \frac{\sum_{i=0}^{k} a_i\, n^i}{n^k} = a_k + \lim_{n\to+\infty} \sum_{i=0}^{k-1} \frac{a_i}{\underbrace{n^{k-i}}_{\to 0}} = a_k \in \mathbb{R}_{>0}$$

- In particular, the polynomial is asymptotically positive, so the limit criterion applies

□

Summary

- A **constant added** to the argument of a power, or a **constant factor** in it, does not change the order of growth:

$$(n + \lambda)^\alpha = \Theta(n^\alpha), \qquad (\lambda\, n)^\alpha = \Theta(n^\alpha)$$

- n^α is super-linear if $\alpha > 1$ and sub-linear if $\alpha < 1$

- A polynomial of degree k with positive leading coefficient is $\Theta(n^k)$: only the leading term counts

Properties of exponential functions

- For exponential functions the picture is different: the base and a constant factor in the exponent **do change** the order of growth
- Only a constant added to the exponent leaves it unchanged

- Any exponential function of a natural variable with a base strictly greater than one is ω of the exponential functions of a natural variable with a smaller base, and o of the exponential functions of a natural variable with a greater base

Observation 13: different bases

For every $b \in \mathbb{R}_{>1}$ and $c \in \mathbb{R}_{>1}$, we have:

- $b^n = \omega(c^n)$, *if* $b > c$;
- $b^n = o(c^n)$, *if* $b < c$.

Proof

For every $b \in \mathbb{R}_{>1}$ and $c \in \mathbb{R}_{>1}$, since $b = c^{\log_c b}$, we have:

$$\lim_{n \to +\infty} \frac{b^n}{c^n} = \lim_{n \to +\infty} \frac{c^{n \log_c b}}{c^n} = \lim_{n \to +\infty} \left(c^{\log_c b - 1} \right)^n$$

$$= \begin{cases} +\infty & \text{if} \quad b > c \\ 0^+ & \text{if} \quad b < c \end{cases}$$

- since $c^{\log_c b - 1} = \frac{b}{c}$ is greater than one if $b > c$ and belongs to $(0, 1)$ if $b < c$ (limit of the geometric progression)



- Multiplying the exponent of an exponential function of a natural variable by a positive constant changes its order of growth, unless the constant is equal to one

Observation 14: scaling of the exponent

For every $b \in \mathbb{R}_{>1}$ *and* $\lambda \in \mathbb{R}_{>0}$*, we have:*

- $b^{\lambda n} = \omega(b^n)$*, if* $\lambda > 1$*;*
- $b^{\lambda n} = o(b^n)$*, if* $0 < \lambda < 1$*.*

Proof

For every $b \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}_{>0}$, we have:

$$\lim_{n \to +\infty} \frac{b^{\lambda n}}{b^n} = \lim_{n \to +\infty} \left(\frac{b^\lambda}{b} \right)^n$$

$$= \lim_{n \to +\infty} \left(b^{\lambda - 1} \right)^n$$

$$= \begin{cases} +\infty & \text{if} \quad \lambda > 1 \\ 0^+ & \text{if} \quad 0 < \lambda < 1 \end{cases}$$



An alternative proof

- An alternative proof is based on the fact that

$$b^{\lambda n} = \left(b^{\lambda}\right)^{n} \qquad \text{and} \qquad b^{\lambda} \begin{cases} > b & \text{if} \quad \lambda > 1 \\ < b & \text{if} \quad 0 < \lambda < 1 \end{cases}$$

- Accordingly, by the Observation «different bases» applied to the bases b^{λ} and b, we have

$$b^{\lambda n} = \omega(b^n) \text{ if } \lambda > 1, \qquad b^{\lambda n} = o(b^n) \text{ if } 0 < \lambda < 1$$

Example

- We have $2^{n/2} = o(2^n)$, since

$$2^{n/2} = \left(2^{1/2}\right)^n \qquad \text{and} \qquad 2^{1/2} = \sqrt{2} < 2$$

- We have $2^{3n/2} = \omega(2^n)$, since

$$2^{3n/2} = \left(2^{3/2}\right)^n \qquad \text{and} \qquad 2^{3/2} = 2\sqrt{2} > 2$$

- Shifting the exponent of an exponential function of a natural variable by a constant does not change its order of growth

Observation 15: shift of the exponent

For every $b \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}$, we have $b^{n+\lambda} = \Theta(b^n)$.

Proof

For every $b \in \mathbb{R}_{>1}$ and $\lambda \in \mathbb{R}$, we have:

$$\lim_{n \to +\infty} \frac{b^{n+\lambda}}{b^n} = \lim_{n \to +\infty} \frac{b^n \, b^\lambda}{b^n} = b^\lambda \in \mathbb{R}_{>0}$$



Constant in the exponent: added or multiplied

- For example,

$$2^{n+10} = 1024 \cdot 2^n = \Theta(2^n), \qquad \text{while} \qquad 2^{2n} = 4^n = \omega(2^n)$$

- A constant **added** to the exponent is a constant factor
- A constant **factor** in the exponent changes the base

Summary

- A **different base** changes the order of growth: $b^n = o(c^n)$ if $b < c$, and $b^n = \omega(c^n)$ if $b > c$
- A **constant factor** $\lambda \neq 1$ in the exponent changes it too, because $b^{\lambda n} = (b^\lambda)^n$ has base b^λ: for example $2^{n/2} = o(2^n)$
- A **constant added** to the exponent does not change it: $b^{n+\lambda} = b^\lambda \, b^n = \Theta(b^n)$

In summary

Introduction: In summary

- We collect the key points of the chapter
- They are the tools to compare the running times of algorithms in the rest of the course

Summary (1/3)

- The functions of a natural variable used in the analysis of algorithms are logarithmic ($\log_a n$), power (n^α), exponential (b^n), factorial ($n!$) and self-power (n^n) functions, together with their sums and products
- The orders of growth are compared with the asymptotic notation of the chapter «Asymptotic notation»
- When the limit of $\frac{f(n)}{g(n)}$ exists it decides (limit criterion): 0^+ gives $f(n) = o\big(g(n)\big)$, a positive real number gives $f(n) = \Theta\big(g(n)\big)$, $+\infty$ gives $f(n) = \omega\big(g(n)\big)$

Summary (2/3)

- Hierarchy (with $a, b > 1$ and $\alpha, \beta > 0$):

$$\log_a^\beta n = o(n^\alpha), \quad n^\alpha = o(b^n), \quad b^n = o(n!), \quad n! = o(n^n)$$

- The base of a logarithm, a constant added to the argument of a logarithm or of a power, a constant factor in the argument of a power, and a constant added to the exponent of an exponential **do not change** the order of growth (the two functions are Θ of each other)

- A polynomial of degree k with positive leading coefficient is $\Theta(n^k)$

Summary (3/3)

- A constant factor in the exponent of an exponential, and a different base of an exponential, **do change** the order of growth: for example

$$2^{n/2} = o(2^n) \qquad \text{and} \qquad 3^n = \omega(2^n)$$

- A function has a **sub-linear**, **linear** or **super-linear** order of growth if it is $o(n)$, $\Theta(n)$ or $\omega(n)$; for example $\sqrt{n}$, $5n + 3$ and $n \ln n$